(1)

A) Computer Network is a collection of interconnected Computers, servers and other devices that communicate with each other through Communication channels such as cables, fiber optics, or wirless signals. The main purpose of a network is to enable sharing of resources and information. Networks allow signals ~~The main pur~~ users to share files, printers, applications, and internet connections efficiently. They also support communication methods like email, video conferencing, and messaging. Computer networks improve productivity, reduce operational costs, provide centralized control of data and allow remote access to information, making them essential in modern organizations, education, and daily life.

Answers

(2) An IP address is a unique numerical identifier assigned to every device connected to a computer network that uses the Internet protocol for communication. It helps identify the source and destination of data packets so that information can be delivered correctly. There are two main version; IPV4, which is a 32-bit address written in decimal format. eg: 192.168.11. and IPV6. which is a 128-bit address written in hexadecimal format to support a larger number of device. IP address can be static (manually assigned and fixed) or dynamic (automatically) assined by DHCP server). They are essential for communicate between devices over network and the Internet.

(3) LAN (Local Area Network) and WAN (wide Area Network) are categories of computer networks based on their geographical coverage. A LAN typically covers a small area such as a home, office or school and usually provides high data transfer speeds and low latency because it is locally managed. It is often owned and maintained by a single organization. In contrast, a WAN spans larger areas such as cities or countries and connects multiple smaller networks together. However, WANs generally offer faster speeds than LANs because they use advanced transmission technologies and are easier to set up and maintain. The Internet is considered a WAN, and unlike LANs, WANs do not reply on services providers for connectivity.

(4)

A router is a hardware device mainly used to store files and run software applications for users on a network.

It is primarly function is to increase the physical speed of internet cables and generate wifi-signals directly from satellites without needing any network connection. Routers operate at the Application layer of the OSI model and do not handle data packets or addresses. They are typically used to replace computers in a network and eliminate the need for IP address. Modern routers are designed mainly for printing documents and charging other electronic devices rather than managing network traffic.

(5)

The OSI model is a networking model that explains how communication happens between devices. It has several layers that help data travel from one system to another and is used to understand network functions.